

Design and development of a smart sunbird feeder for autonomous video-based behavioral monitoring



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Abstract

In the ever-growing field of automated bird monitoring and identification, the range of cost-efficient and non-invasive methods for ecological monitoring of species of birds is wanting. The aim of this project is to investigate the design and implementation of a smart sunbird feeder capable of detecting and identifying nectar-feeding birds using an embedded ML algorithm coupled with a computer-vision system while remaining affordable. The system used was trained to recognize 22 species of birds with a focus on the 6 nectar-feeding species known to reside in the Cape Town area. Across these species, the system achieved an accuracy of ___ in identifying the species correctly and achieved an average inference time of _____. These results demonstrate the feasibility of using a targeted, low-cost smart feeder as an automated monitoring system and provide a basis for more efficient collection of species-level visitation data in future ecological studies.

Acknowledgements

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Chapter 1

Introduction

1.1 Background

Birds are integral members of our ecosystems as providers of many roles that prove vital to the health of these ecosystems, be they predators, pollinators, scavengers, seed dispersers, seed predators, or ecosystem engineers [1]. Sunbirds in particular have proven to be especially important in the pollination process and are a dominant group of nectar feeders throughout Africa. In the Western Cape specifically, sunbirds have been found to be the primary pollinators of several plant species [2]. Furthermore, in Cape Town nectarivorous birds have been established to play an unusually important role in the ecosystem due to the highly asymmetrical mutual pollination between birds and plants in this biodiversity hotspot [3]. Despite their integral ecological role, however, the capturing of data of when, how often and what species of this group interact with particular food sources remains a challenging task. The current means of obtaining these measurements are time-consuming and are heavily dependent on human labour. These factors have led directly to practical limitations on the monitoring of these birds.

More recently, camera traps have become a much more efficient means of remotely capturing animal distribution, abundance and behaviour, but this method of ecological monitoring remains limited by sampling errors such as false triggers and data extraction from this footage is still an expensive, time-consuming and manual task [4, 5]. With advancements in the fields of digital imaging, embedded computing and computer vision, automatic and remote ecological monitoring becomes a much more feasible option while simultaneously reducing the amount of false detection instances [6]. Already, machine learning has been proven able to identify animals from camera trap footage with a highly reliable accuracy [5]. Such developments make possible a compact and cost-effective method of not only collecting data for ecological monitoring, but doing so intelligently with minimal or no requirement of human intervention.

The monitoring of nectarivorous birds is made much more viable at known hotspots of nectarivore activity such as feeding sites. Introducing a nectar feeder to a nearby feeding location is a sound method of localizing bird activity to a well-constrained area as well as standardizing an angle of captured footage, since the feeding port requires these birds to perch on a specific platform. Recent studies have proven the viability of embedded machine learning algorithms in detecting and monitoring avian sightings at feeder stations [7]. An adapted system for nectar-feeding birds that similarly incorporates video monitoring, automated feeder visit detection and species identification could prove to contribute significantly to the available monitoring methods of these birds.

1.2 Problem Statement

Sunbirds are difficult creatures to monitor. They are small, highly mobile and easily disturbed. Current aviary monitoring methods have shown that a combination of camera recording, embedded computing and machine learning makes automated monitoring viable.

Analysis of current projects expose an engineering gap that this project will aim to fill. There is a need for a compact and reliable system designed to cater specifically to nectar feeding birds and their physical behaviour while being capable of unsupervised operation, reliable footage capture with associated species predictions and sufficient off-grid capacities.

Existing Technology includes:

- Camera-equipped smart feeders
- Raspberry Pi/embedded wildlife-monitoring systems
- Motion/event-triggered image or video capture
- Machine-learning species classification
- Solar-powered monitoring systems

Limitations of existing technology are:

- Designed around seed feeders or hummingbird-style products
- Dependent on external/cloud infrastructure in some cases

- Not specifically designed around local South African nectar-feeding birds
- Not optimised for low-power autonomous field operation

Considering this engineering gap in the field of automated bird monitoring as well as the relevant existing technology and its limitations, the following problem statement for this project was defined:

1.3 Objectives

This project aims to design and deploy an autonomous and remote smart nectar feeder, with a primary focus on monitoring sunbirds in the Cape Town area. The system will utilise a machine learning algorithm locally embedded on an edge processing device to classify classes of interest and record visits of such classes. These recordings will be made available to users via a locally-hosted web interface.

For the objectives of this project to be achieved, a functional prototype of the complete system must be developed, assembled and tested in the field. The objectives this project aims to achieve through the realisation of a successful prototype include:

- Designing and constructing a nectar feeder arrangement suitable for sunbirds and camera monitoring
- Developing an automated system for detecting bird visits
- Capturing video footage of detected visits
- Implementing embedded species classification using a locally relevant dataset
- Storing video recordings and their associated observation metadata locally
- Developing an autonomous power system using battery and solar energy
- Integrating the subsystems into an deployable prototype
- Evaluating detection, classification, video, power and overall system performance

1.4 Scope and Limitations

The scope of this project defines the functional and technical boundaries within which the smart sunbird feeder will be designed, implemented and evaluated. These boundaries

are intended to keep the project focused on the development of a reliable and practical monitoring system, while acknowledging constraints associated with time, resources, deployment conditions and system capability. The scope of this project is set as follows:

- **Hardware scope:** Data collection will use a Raspberry Pi, a single camera module and a single Time-of-Flight (ToF) sensor.
- **Time scope:** Monitoring will be limited to daytime bird activity.
- **Subject scope:** The system will focus on sunbirds and other relevant nectar-feeding bird species.
- **Monitoring scope:** Bird visits will be recorded using motion-triggered capture rather than continuous monitoring.
- **Location scope:** System deployment and evaluation will be limited to Cape Town.
- **Data scope:** Captured data will consist of a short video recording, the predicted species and corresponding confidence score, and the date and time of the observation.
- **Storage scope:** All captured data will be stored locally on the system.
- **Ethics scope:** Species classification and monitoring will be conducted using entirely non-invasive methods.

Similarly, the limitations of this project are defined as follows:

- **Hardware limitation:** The project is limited to adapting a commercially available nectar feeder and integrating existing electronic components. Custom PCB design and development of new circuitry will not be undertaken.
- **Species limitation:** Only bird species relevant to South Africa will be considered as classes of interest. Foreign species will not be included.
- **Classification limitation:** Classification will be limited to species level. Sex, age and individual identification will not be considered.
- **Data collection limitation:** Bird behaviour will not be analysed. Collected data will be limited to species identification, the date and time of each visit, and an accompanying video recording.
- **Time limitation:** The project must be completed by 26 October 2026.
- **Budget limitation:** The project is constrained to a budget of R2000.

- **Energy limitation:** Solar-energy generation is dependent on environmental conditions, particularly solar irradiance and cloud cover.
- **User interface limitation:** User interaction with collected data will be limited to a simple, locally hosted web interface. Development of a dedicated desktop or mobile application will not be considered.

1.5 Thesis Outline

The remainder of this thesis is organized as follows:

Chapter 2, Background: Background and literature review

Chapter 3, Methodology: Project methodology

Chapter 4, Design: Design and development

Chapter 5, Results: Experimental results

Chapter 6, Conclusions: Conclusions and future work

Chapter 2

Literature Review

2.1 Introduction

The purpose of this literature review is to survey and investigate relevant literature to aid the design of a smart sunbird feeder. The literature reviewed will inform subsequent design choices and methods. Literature concerning ecological and avian monitoring will support the motivation for research in this area, while identifying sunbird behaviours and characteristics of nectar feeders that may present difficulties or limitations. Existing automated bird-monitoring solutions will be analysed to determine how aspects of similar research can best be applied. Several key enabling technologies will then be investigated, together with their respective advantages and disadvantages. Finally, the literature review will identify important trade-offs in similar projects and address the research gap this project aims to fill. The findings will support informed design and implementation decisions based on the accomplishments and limitations reported in the literature.

2.2 Bird monitoring and nectarivorous ecology:

Firstly, it is important to establish the relevance of ecological monitoring as motivation for research in this field, which this section aims to accomplish.

Bird activity is a vital metric for determining ecosystem biodiversity and functioning. Nations have adopted methods of observing this activity to support sustainable development strategies and gauge environmental and ecosystem health [8]. Conventional avian ecological monitoring includes approaches such as human point counts, passive sound recording and camera traps. These methods have proven practically identical in effectiveness when implemented correctly, while remote observation is preferred for remote deployment or rare species monitoring [9]. Sound recording and similar autonomous methods reduce the time required for human monitoring, provide permanent data capture and eliminate

2.2. BIRD MONITORING AND NECTARIVOROUS ECOLOGY:

human observer bias. These methods also have drawbacks, including equipment and storage costs and the risk of data loss if they fail [10]. Despite these shortcomings, remote monitoring systems, particularly camera traps, are widely used for wildlife monitoring. Depending on the model, set-up and application, camera traps are viable for a diverse range of scenarios [11]. **The advantages, limitations and project suitability of these monitoring methods are summarised in Table 2.1.**

Table 2.1: Comparison of common methods used for monitoring bird activity.

Method	Principal advantages	Principal limitations	Suitability for this project
Human point counts	Can identify birds using both visual and acoustic observations; requires relatively little equipment; experienced observers can interpret behaviour and surrounding conditions immediately.	Requires trained observers and considerable field time; observations are restricted to scheduled survey periods; susceptible to observer bias; generally provides no permanent visual record.	Low. Useful for manually validating system observations, but unsuitable as the primary method for continuous autonomous monitoring.
Passive acoustic monitoring	Enables long-duration unattended recording; provides a permanent record; can detect birds that are hidden from view; reduces the need for an observer to remain in the field.	Produces large volumes of audio data; performance is affected by background noise and overlapping calls; silent birds cannot be detected; reliable species identification may require specialist analysis or automated classifiers.	Moderate. Valuable for monitoring vocal activity, but it cannot provide the visual information required by the proposed image-based classification system.
Camera-trap monitoring	Provides permanent visual records; enables verification of species, visit duration and aspects of behaviour; can operate autonomously; supports automated image or video analysis.	Detection depends on camera placement and trigger performance; false and missed activations may occur; image quality is affected by lighting, motion and occlusion; video requires substantial storage and energy.	High. The predictable feeding location allows the camera field of view and trigger region to be constrained, making this the most appropriate primary monitoring method.

Sources: Comparison informed by Darras et al. [9], Shonfield and Bayne [10], and Trollet et al. [11].

In Cape Town, sunbirds have been shown to play a particularly important ecological role, with the four local nectar-feeding species contributing significantly to the maintenance of the region’s many bird-pollinated plant species [12]. This role is accompanied by behavioural traits that affect how they should be monitored. Although sunbirds can hover while feeding, they frequently perch near nectar sources when an appropriate platform is available. Providing a perch therefore makes camera-based monitoring more practical [13]. Sunbirds also associate food-source quality with colour, making feeder appearance more important than for seed-feeding species [14]. Many nectar feeders therefore use red feeding ports because red is common among flowers, highly visible in natural settings and less detectable to bees, reducing competition at the food source [15]. Sunbird feeding routines also adapt to the location and quality of available food sources. Feeder activity should therefore not be assumed to represent overall sunbird activity in the surrounding area [16]. While some studies suggest supplemental feeding may concentrate sunbird activity and affect surrounding pollination networks, others report negligible effects on activity at natural food sources [16, 17].

Sunbird feeders differ significantly from ordinary seed feeders because these birds are nectarivorous. Their diet requires feeders to contain liquid rather than solid food and, due to their long bills and "tube-like" tongues, they prefer narrow feeding ports over the trays commonly used in seed feeders [18]. The nectar solution consists of water supplemented with sugar, particularly sucrose. Studies of sunbird feeder preferences found sucrose-rich solutions to be more popular than hexose-rich alternatives [19]. However, storing nectar in feeders can promote microbial transmission and fermentation, resulting in ethanol production. An analysis of hummingbird feeders found non-pathogenic bacteria and fungi that differed significantly from those found in flowers [20]. Additionally, an ethanol concentration of only 2% was found to significantly reduce sunbird visitation [21]. For these reasons, feeder nectar should be replaced regularly.

2.3 Analysis of existing smart bird feeders

This section of the literature review will be dedicated to investigating the efforts of other individuals in similar fields of research and acknowledging the benefits and drawbacks for each of their respective approaches to conducting their research.

2.3.1 Research and experimental smart-feeder systems

Rapid advances in Machine Learning (ML) and the increasing availability and affordability of microcomputers such as the Raspberry Pi (RPi) have enabled the development of smart bird feeders by both companies and academics. Researchers have demonstrated the feasibility of using an RPi as the computational core of a smart seed feeder, reliably capturing data from visiting birds using video, microphone and strain gauge recordings [22]. The system also performed local species inference using an embedded ML algorithm and uploaded data to an independently hosted website for user access and download. However, this approach requires an internet connection for data access, limiting deployment in remote areas. The system also depends on an external power supply, further restricting remote deployment [22].

Another approach to smart bird feeders is to decentralise the computational load by relying on a remote server. By moving computation away from the physical feeder, researchers were able to monitor an unmodified, commercially available seed feeder using a standard internet-enabled camera trap. The Artificial Intelligence (AI) used for identification was also trained only on bird species relevant to the area, substantially reducing computational demand compared to recognising all bird species. Like the previously reviewed feeder, however, the system requires both an internet connection and an external power supply. Although narrowing the training database reduced computational demand, field accuracy fell from 99.5% to 88%, emphasising the importance of representative training data [23].

When shifting focus from seed-feeding to nectar-feeding birds, monitoring approaches often rely on radio-frequency identification (RFID) rather than computer vision and ML. Although RFID enables highly reliable identification, implementation is invasive because tags must be physically attached to the birds. One smart seed feeder using RFID relied on an RPi [7], while another used a simpler Arduino microcontroller. Because RFID does not require an ML component, the computing power required to operate the system can be significantly reduced [24].

2.3.2 Commercial smart nectar feeders

Smart bird feeders are not exclusively academic projects, with several commercially available versions now on the market. The Birdbuddy Smart Hummingbird Feeder is one

relevant example [25]. One notable design feature is its colour choice: the feeding port area is made from translucent red plastic, aligning with earlier literature showing that nectar-feeding birds associate food sources with colour. Red is also the most common colour used by feeder manufacturers, further supporting this literature. Another commercial example is the Birdfy Hum Feeder, which has a very similar design [26]. Both products depend on proprietary software, cloud-based storage, a mobile application and consistent internet access. The principal characteristics, advantages and limitations of the academic and commercial smart bird feeders reviewed are summarised in Table 2.2.

2.3.3 Comparative findings and design implications

An investigation of existing smart bird feeders, developed for either research or commercial purposes, reveals several recurring design choices and limitations. Although most available literature concerns seed feeders, smart nectar feeders have also been developed. Most systems rely on a microcomputer, typically an RPi, rather than a microcontroller to implement ML-based, non-invasive species identification. Another common design choice is to offload data storage and user interaction to an external or cloud-based server rather than hosting locally. This reduces local computational overhead but requires an internet connection. Nearly all inspected systems also rely on a constant external power source, severely limiting deployment in remote areas.

Table 2.2: Comparison of selected academic and commercial smart bird-feeder systems.

System	Feeder	Monitoring	Processing and storage	Power	Key limitation or lesson
Niers et al. [22]	Seed	Video, audio and strain-gauge measurements.	RPi performs local species classification and uploads results to a hosted website.	External supply.	Demonstrates non-invasive multimodal monitoring, but requires network access and external power.
Mansouri [23]	Seed	Motion-triggered IP camera recording short videos.	Videos are transferred to a local server for bird detection and classification of 40 local species.	Off-grid supply not specified.	Localised training reduces computational demand, but field accuracy fell from 99.5% to 88%.
Youngblood [7]	Seed	Perch-mounted RFID detects tagged birds.	RPi Zero W stores tag identifiers and timestamps.	Portable battery.	Enables reliable individual identification, but requires invasive tagging and cannot identify untagged birds.
Ibarra et al. [24]	Nectar	RFID detection with servo-controlled feeding port.	Microcontroller controls access and stores tag identifiers and timestamps on an SD card.	Two 9 V batteries.	Enables low-power local operation, but requires invasive tagging and provides no visual identification.
Birdbuddy Smart Hummingbird Feeder [25]	Nectar	Automatic photo and video capture.	Proprietary identification software with recordings accessed through a mobile application.	Rechargeable battery; optional solar.	Polished consumer system, but depends on proprietary software and network connectivity.
Birdfy Hum Feeder [26]	Nectar	Motion-triggered 1080p video and automated identification.	Recordings and results are accessed through a proprietary application.	Rechargeable battery; optional solar.	Demonstrates solar-supported monitoring, but depends on proprietary software and connectivity.

2.4 Technology behind autonomous bird monitoring systems

This section will refine upon the broader topic of existing smart feeders discussed in the previous section and will focus on several of the technologies that comprise the functional pipeline of such feeders. The design choices behind the technology at each stage of this pipeline will be investigated in order to better inform the design of this project. The principal stages of a generic autonomous bird-monitoring pipeline are illustrated in Figure 2.1.

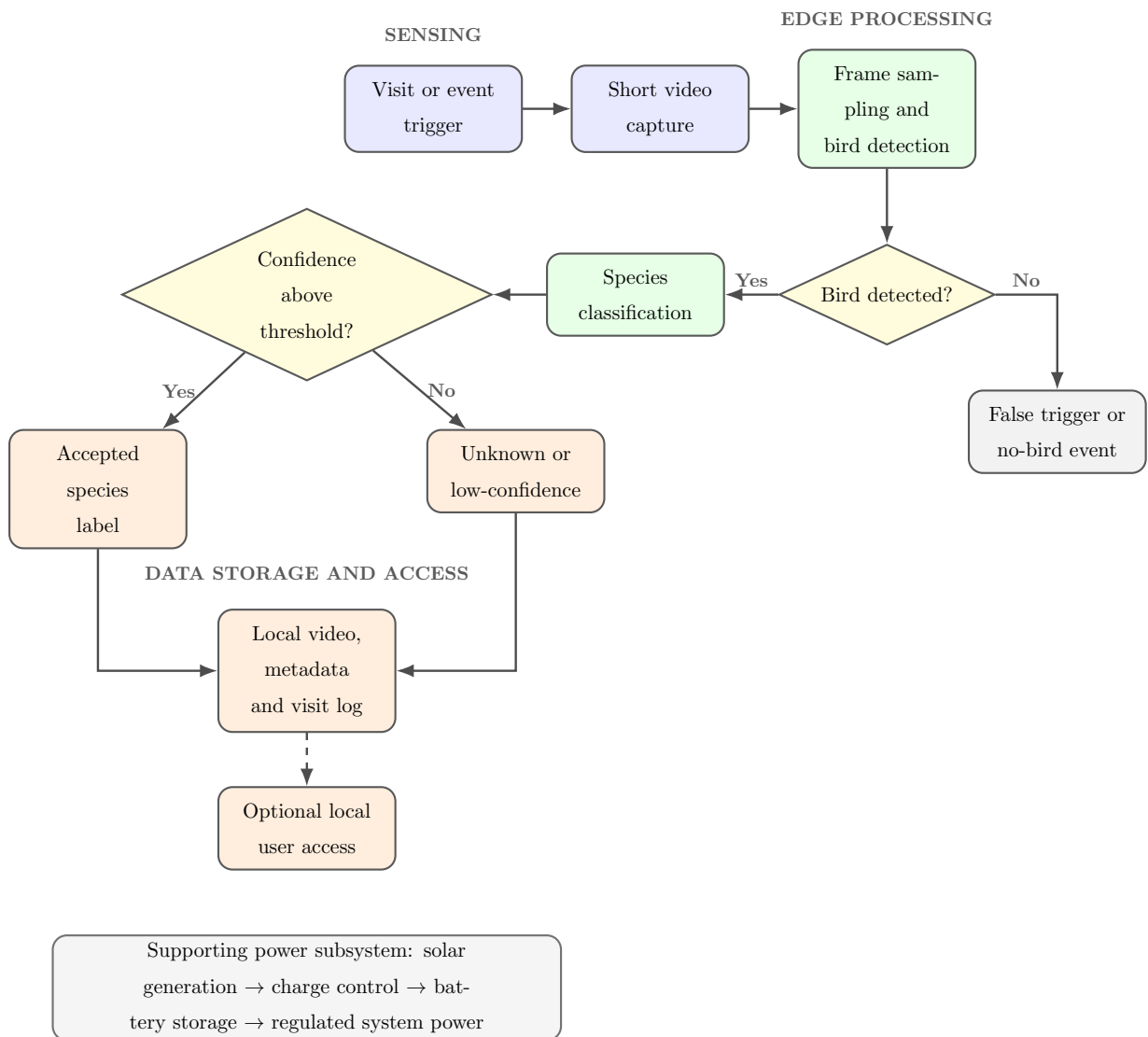


Figure 2.1: Generic functional pipeline for an autonomous, camera-based bird-monitoring system. Solid arrows represent the primary data path, while the dashed arrow represents optional user access.

2.4.1 Camera-based monitoring

Although acoustic recording is regarded as a reliable means of remote bird monitoring, this project will instead use a camera-based system. As most existing smart bird feeders similarly use this medium, its benefits and drawbacks should be investigated.

As this project aims to provide non-invasive species identification using computer vision and ML, an important design decision is whether to record video or still images. Still images require significantly less storage and power [27, 28], while video provides a richer user experience and is associated with improved species-identification accuracy [27].

Video recording is less efficient in terms of both energy and storage, influencing whether the system should monitor continuously or use a trigger mechanism. Given the project's dependence on solar power, continuous camera monitoring would impose excessive energy requirements, so a triggered system will be implemented [28]. Although triggering significantly reduces power consumption, it introduces reliability concerns, including false triggers of an empty feeder and missed visits. Calibration and field testing must therefore be prioritised to minimise these events [29].

Having reviewed some of the means of monitoring bird remotely, a comparison of them can be conducted and their trade-offs evaluated. The principal trade-offs associated with media format and capture activation are summarised in Table 2.3.

Table 2.3: Trade-offs between alternative media formats and capture strategies for autonomous bird monitoring.

Option	Principal advantages	Principal limitations	Suitability for this project
<i>Media format</i>			
Still images	Require relatively little storage and processing; individual images can be captured and classified quickly; impose a lower power demand than video.	Provide only one viewpoint per capture; may contain motion blur, occlusion or an unsuitable pose; cannot directly record visit duration or movement.	Moderate. Suitable for basic identification, but a single image may not provide sufficient evidence for reliable classification.
Short video clips	Provide multiple frames and viewpoints; support selection of the clearest frame; record visit duration and offer a more complete visual record.	Require more storage, processing time and energy than still images; may produce redundant or unusable frames.	High. The additional visual information improves the likelihood of recording a useful view while keeping resource use manageable.
<i>Capture activation</i>			
Continuous capture	Minimises dependence on trigger performance; provides complete coverage of activity within the camera field of view.	Produces large volumes of irrelevant footage; imposes high storage, processing and power demands; reduces battery-supported operating time.	Low. The resource demand conflicts with the requirements for portable, solar-powered operation.
Event-triggered capture	Records media only when activity is detected; substantially reduces unnecessary storage, processing and energy use.	Performance depends on trigger sensitivity and positioning; delayed, missed or false activations may occur and must be evaluated experimentally.	High. It provides the most practical balance between visit coverage and resource consumption for autonomous deployment.

Design implication: Short, event-triggered video clips provide the most suitable combined strategy for the proposed system. This approach retains the principal identification benefits of video while limiting the storage and energy demands associated with continuous recording.

2.4.2 Automated species detection

Although rapid advances in ML and computer vision have made automated species identification increasingly feasible, several implementation decisions remain.

Such algorithms can achieve accuracies comparable to humans when implemented correctly, with researchers reporting 99.3% accuracy for mammal identification [30]. Similar results have been obtained for birds, with one study reporting 96.71% accuracy across 10 species [31].

Bird identification using computer vision relies on fine-grained image classification (FGIC). Unlike conventional image classification, which distinguishes between distinct categories such as a ball and a dog, FGIC distinguishes between visually similar subcategories, such as dog breeds [32]. FGIC is particularly sensitive to insufficient training data, but institutions and crowd-sourced initiatives have made large labelled datasets such as ImageNet and NABirds publicly available [33].

Despite high benchmark accuracies, model performance can deteriorate substantially when algorithms trained on idealised online datasets are deployed in the field. Error rates have been found to increase by 92% to 115% under field conditions [34]. This highlights the importance of incorporating geographically relevant training data.

Public bird datasets can be extensive. Birds 525 contains nearly 90 thousand labelled images across 525 species [35], while Semi-Aves contains over 160 thousand images across 1 000 species [36]. The image and class scales of several commonly used bird-identification datasets are compared in Figure 2.2. However, narrowing the training scope can improve performance. One study found that removing the rarest species increased identification accuracy from 72.8% to 76.3% [37]. Using a smaller, locally relevant class set is also common in automated bird-identification studies [23, 37]. The model used in this project will therefore be limited to bird species relevant to Cape Town.

A smaller class set introduces the challenge of handling unidentified birds and other off-target detections. Two scenarios must be considered: low-confidence predictions and classes unknown to the model. Low-confidence predictions can be handled using selective classification, or a "reject option", in which predictions below a confidence threshold are rejected [41]. Unknown classes arise because the model operates in an open-set environment, where it may encounter classes absent from its training data [42]. A similar rejection mechanism should therefore be used to prevent unreliable classification of unknown visitors.

2.4.3 Embedded and autonomous usage

Given that this project aims to rely on solar power for continuous remote operation, literature concerning its embedded and autonomous functionality must also be reviewed.

While most existing smart feeders discussed previously rely on external servers or cloud services for data storage and processing, this project will instead perform these functions

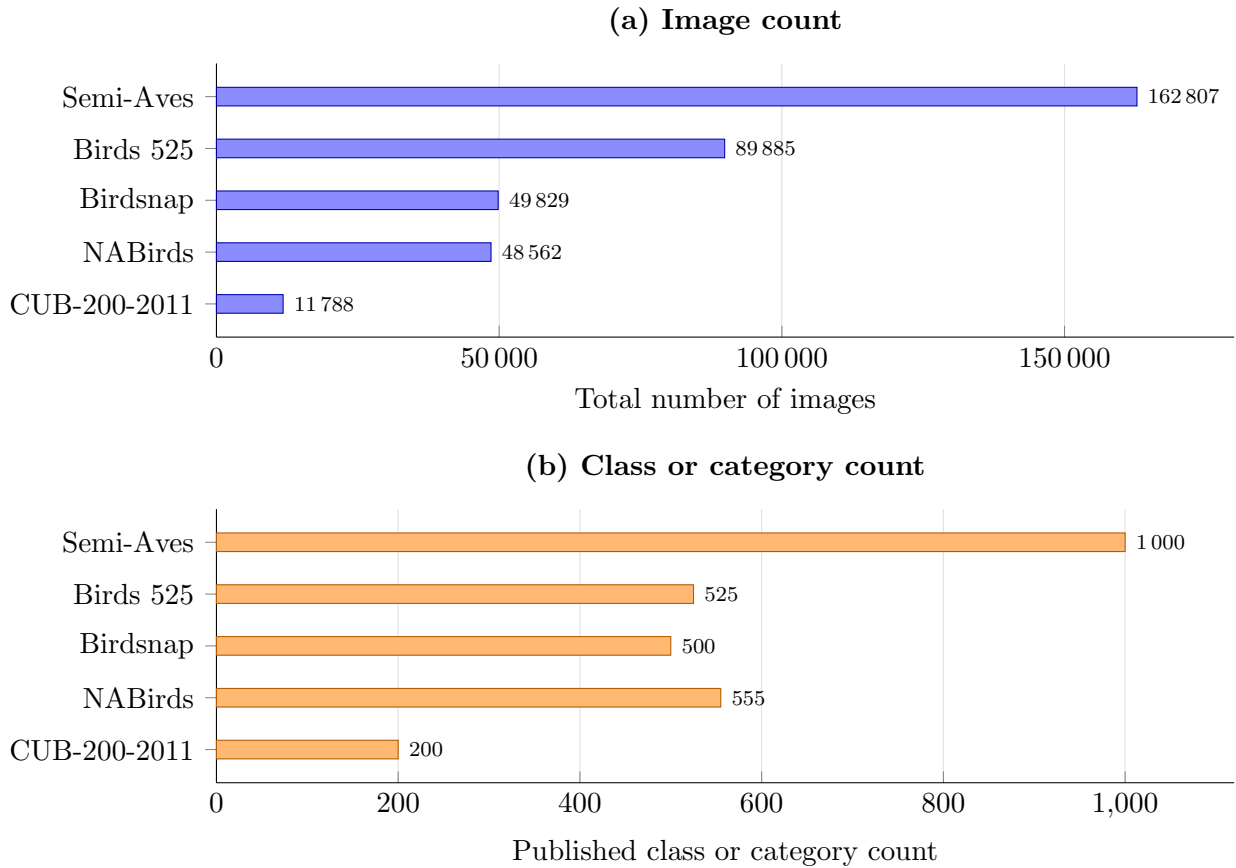


Figure 2.2: Scale of selected bird-image datasets by total image count and published class or category count. Adapted from dataset statistics reported by Wah et al. [38], Van Horn et al. [33], Berg et al. [39], Gpiosenka [40], and Su and Maji [36].

on-device using edge processing. Besides removing dependence on remote processing, edge processing enables immediate decisions on whether footage should be recorded, reducing storage of irrelevant data [43].

This project requires an inference model capable of identifying bird visits quickly enough to determine whether recording should occur. Microcontrollers offer low-power operation and are well suited to simple sensing and processing, but their limited architecture restricts demanding tasks such as inference and image classification. Consequently, these tasks generally take considerably longer than on a microcomputer, limiting the suitability of microcontrollers for neural-network applications [44].

Microcomputers provide greater processing capability but require significantly more power while operating. Trigger-based recording can reduce storage requirements, but will only similarly reduce power consumption if components, or the system as a whole, can enter and exit an idle state when inactive [45]. The principal differences between conventional

microcontrollers and Raspberry Pi-class single-board computers are summarised in Table 2.4.

Table 2.4: Comparison of conventional microcontrollers and Raspberry Pi-class single-board computers.

Criterion	Microcontroller	Raspberry Pi-class SBC	Project relevance
Power	Very low consumption with effective sleep modes.	Typically consumes several watts when active.	Microcontrollers offer better low-power operation.
Start-up	Begins operating almost immediately.	Requires several seconds to boot.	Frequent SBC power cycling may delay capture.
Memory	Usually kilobytes to a few megabytes.	Commonly provides several gigabytes.	Video processing requires SBC-level memory.
Camera and video	Limited resolution and video-handling capability.	Supports high-resolution cameras and sustained video capture.	Short-video recording favours the SBC.
Machine learning	Restricted mainly to compact TinyML models.	Supports larger models and inference accelerators.	Species classification favours the SBC.
Software	Runs embedded firmware or a real-time operating system.	Runs Linux with established camera and ML libraries.	The SBC simplifies system integration.
Data and web access	Storage, databases and web services are constrained.	Supports local storage, databases and web servers.	Required for visit logging and the web interface.
Reliability	Simple software stack and rapid recovery.	Requires storage, thermal and shutdown management.	SBC reliability must be addressed in the design.
Recommended role	Optional trigger or power-management controller.	Primary system processor.	Add a microcontroller only if testing demonstrates a need.

For a system to operate continuously using only solar power, the usable solar energy generated must meet or exceed the system’s energy demand and associated losses:

$$E_{\text{solar, usable}} \geq E_{\text{system}} + E_{\text{losses}}$$

Although most smart feeders reviewed previously rely on external power, commercial products such as the Birdfy and academic studies demonstrate that solar-powered operation is feasible [26, 46].

Having investigated the key technologies supporting the proposed system, a clearer basis for the design phase has been established. The advantages and limitations of each technology have been considered to identify the options most suitable for this project.

2.5 Conclusion

With the relevant literature investigated, several conclusions can be drawn to support the motivation for this research and inform the design and methods of the proposed system.

Analysis of existing smart bird feeders provides insight into existing solutions and opportunities for improvement. Many systems use neural-network classification to automate non-invasive species identification. However, several rely on external processing, cloud services or network connectivity, limiting their suitability for remote deployment. Similarly, fully autonomous solar operation is not consistently demonstrated. These limitations support the use of local processing, storage and power generation in this project.

The review also identified relatively little research on smart nectar feeders, with most literature focused on seed feeders despite several commercial hummingbird feeders. Focusing on nectar-feeding birds introduces additional design considerations, including feeder colour, nectar replacement and protecting electronic components from nectar spillage and environmental exposure.

In conclusion, the literature reviewed proves the feasibility of the required technology for this project, but these proofs are hardly realised as a combined and complete smart nectar feeder for sunbirds. The findings from the literature support the use of triggered video capture, a locally relevant classification model, edge computing and local storage, together with autonomous solar power. The basis for the system requirements and design decisions developed in the following chapters will be constructed using these findings.

Chapter 3

Methodology and Design

3.1 Introduction

This chapter will be dedicated to describing the methodology used in designing and realising this project, as well as detailing the design choices and process. The first section of this chapter will deal with the former of these two topics.

3.2 Project Methodology

The goals and requirements of this project were stated in Chapter 1 of this report. In Chapter 2 I investigated the methods used by others to accomplish similar goals and meet similar requirements. Using this knowledge and input from my supervisor, I have devised an appropriate methodology to implement when undertaking this project. This methodology caters toward the specific nature of this project (many subsystems, hardware + software, etc), as well as the many constraints around this project (budget, time, etc).

Okay, so basically the method outline is:

1. Identify problem
2. Set goals and requirements
3. Research literature and similar existing projects
4. Decide overall system design
5. Decide on what hardware and software to use
6. Design and test each subsystem
7. Assemble complete system and test
8. Look for and make improvements where possible

9. Cross-reference complete system against initial goals and requirements
10. Measure and remark on results
11. Identify future work and system's shortcomings, if applicable
12. Prioritise something that works over something that would be cool if it worked

3.3 Requirements Definition

Again, the aims and goals of this project were stated in Chapter 1. As an engineer, it is important to translate these sometimes vague/broad points into technical requirements. Using the aims and goals in Chapter 1 as an introduction, I developed the following requirements for the final system: These requirements outline the appropriate final specifications needed for the system.

3.3.1 Functional Requirements

- The system must detect any significant motion at the feeder port
- The system must trigger the classification algorithm once motion is detected
- The system must categorize detected subjects into a defined list of species
- The system must manage low-confidence or unrecognised classes appropriately
- The system must record a video if the classification algorithm determines the source of motion to be of a class of interest
- The system must store any recorded footage in a non-volatile storage medium
- The system must charge from a solar power source utilise a battery for energy storage
- The system must make recorded footage accessible from a self-hosted web page for users to interact with
- The system must utilize an 'energy-saving' mode when idle

3.3.2 Non-Functional Requirements

- The system's feeder must be easily accessible and refillable

- The system must be able to classify within x seconds of motion detection
- Recording of a visit must begin y seconds after motion detection trigger
- Recorded footage should meet a minimum resolution/frame-rate requirement sufficient for classification
- Classification accuracy must be at least $z\%$
- False positive events must be less than $a\%$
- User interaction with the system's web interface must be intuitive, easy and well-documented
- The total Bill of Materials must amount to less than R_+
- The system must have an uptime of at least $_$ hours per day, under full sunlight
- The electronic subsystem must be protected from environmental elements by the system's housing
- The system should not present sharp edges, trapping hazards or harmful contamination risks to birds
- Electronics must remain within their allowable operating temperature range during deployment
- The project must contain sufficient design and build documentation

3.4 Acceptance Testing Procedures

Explain the experimental approach used to determine whether the completed system satisfied the requirements established in Section [3.3](#).

Main categories of system evaluation:

- Mechanical and feeder performance: Verify feeder stability, enclosure protection, and ease of refilling and cleaning.
- Visit detection performance: Measure successful detections and false-positive and false-negative rates against the acceptance limits.
- Video capture performance: Verify that visits are sufficiently recorded and that image quality, focus, frame rate and camera positioning meet requirements.

- Data storage and logging: Verify that videos and metadata are stored correctly, retained after restart or power loss, and that storage remains reliable.
- Classification performance: Measure classification accuracy, false-identification rate, inference time, per-class performance where relevant, and low-confidence rejection performance.
- Web interface performance: Verify access to visit records, classification results and video playback, and confirm correct operation of required functions.
- Embedded-system performance and reliability: Run the system continuously for a defined period and record crashes, missed events, storage errors or interruptions. Verify automatic recovery where required.
- Power consumption and autonomy: Measure power use during idle, recording and classification, and determine battery runtime from a full charge.
- Solar charging performance: Measure whether the solar subsystem can recharge or sustain the battery under representative sunlight conditions.
- Overall integrated-system performance: Verify that visits are detected, recorded, classified, stored and made available to the user without manual intervention, and record any subsystem-integration failures.

3.5 Design Decisions

Introduction to overall system, with basic block diagram of all the interacting subsystems. Explain the pipeline of operation briefly.

3.5.1 Hardware Design Decisions

Microcomputer Reiterate that a microcontroller is inadequate for the inference speed required for this project. Basically, besides the fact that we already had it, the RPi 4B was chosen because:

- It is the least computationally-expensive option that can get the job done
- It is also one of the more budget-friendly options that can get the job done
- Unlike its more expensive counterparts, the 4B has onboard FHD video encoding, which is much less power intensive when recording and storing videos

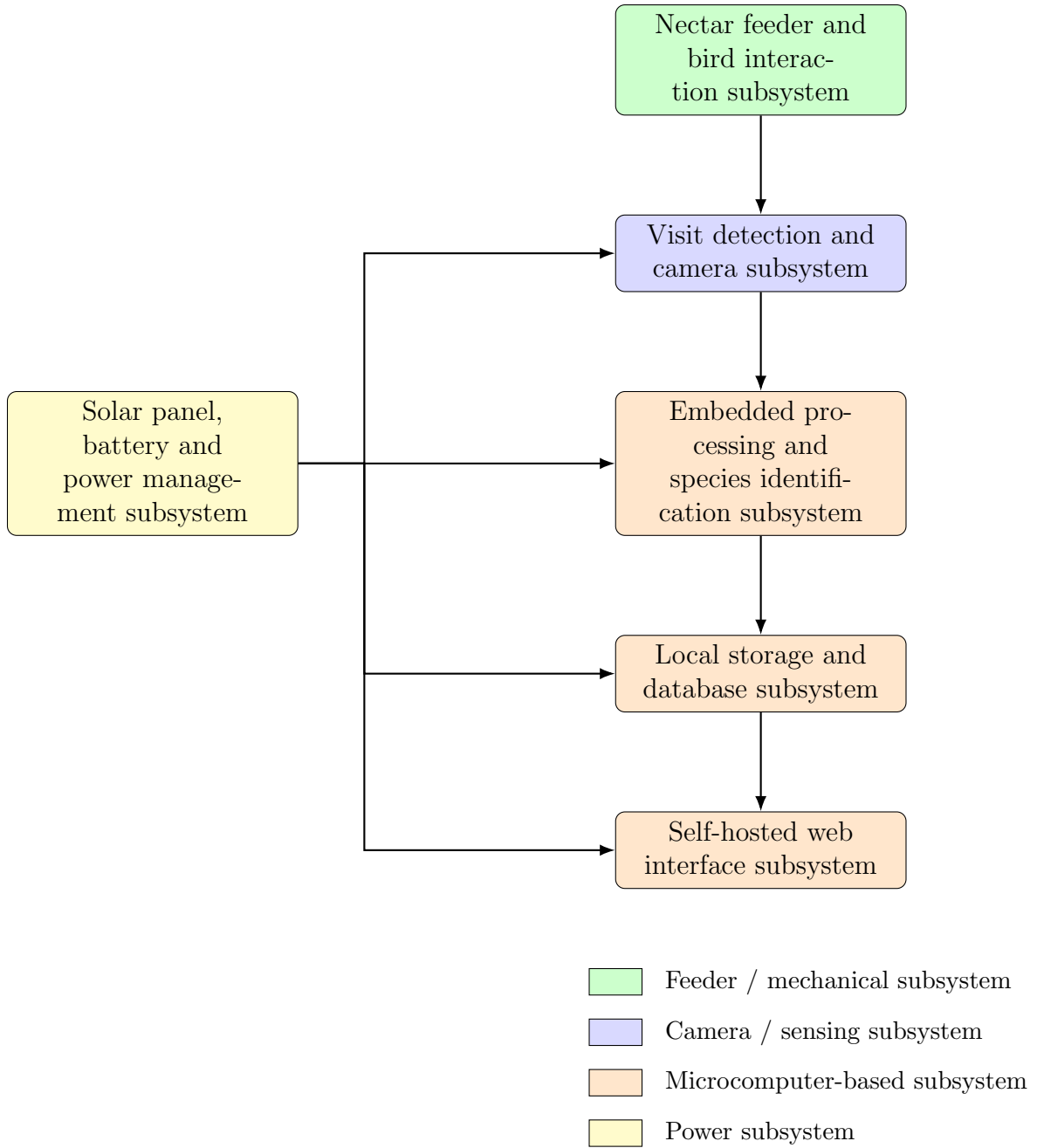


Figure 3.1: High-level block diagram of the proposed smart sunbird feeder system. Blocks are coloured according to the primary hardware subsystem on which they rely.

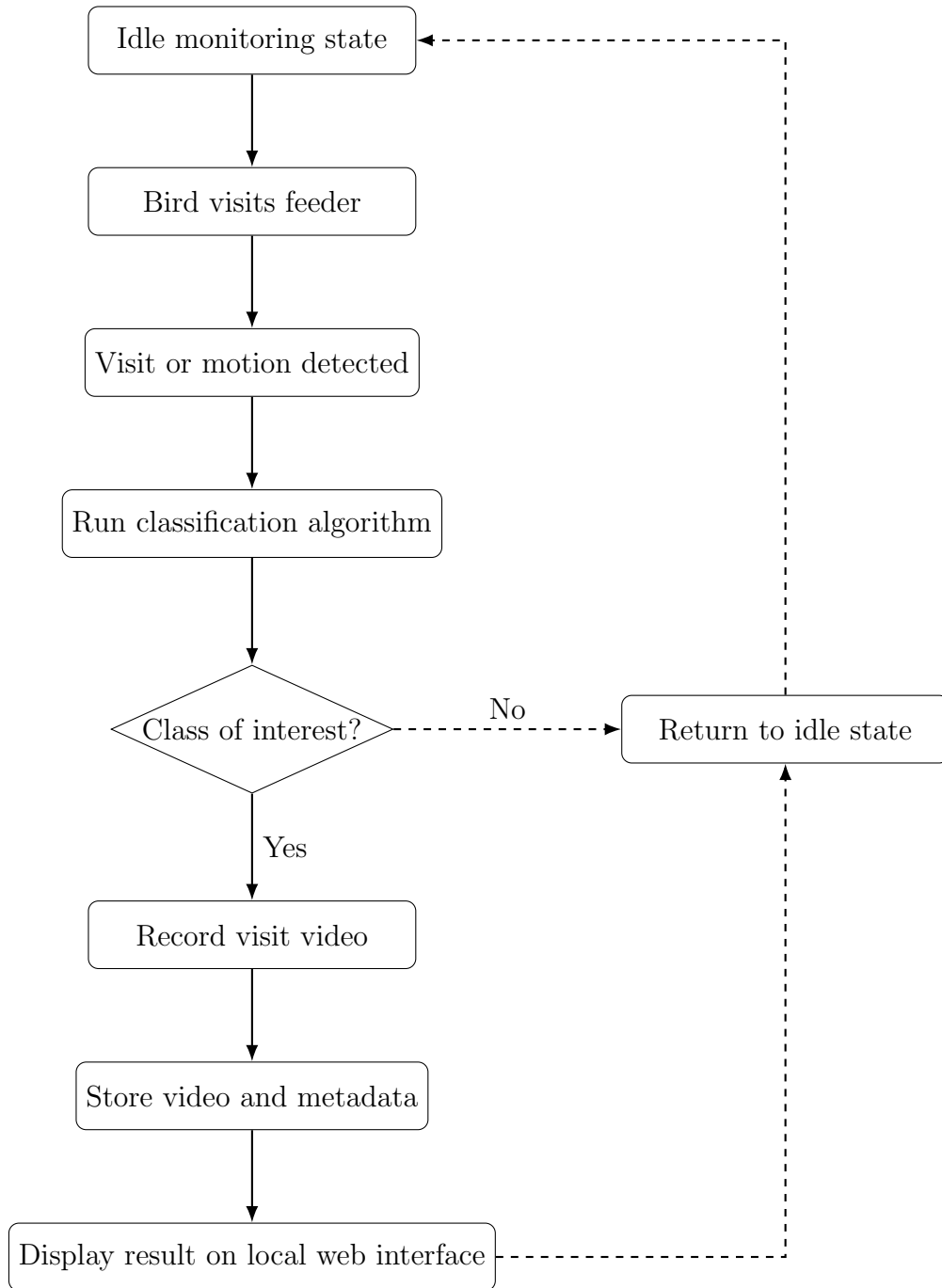


Figure 3.2: Operational pipeline of the proposed smart sunbird feeder system.

- It was already available and has been proven to work well for a similar application, so less design risk and lead time

The candidate platforms were compared according to processing capability, camera compatibility, software support, energy consumption and integration risk. Table 3.1 shows that the Raspberry Pi 4B provides the most appropriate balance for the project.

Table 3.1: Comparison of candidate processing platforms

Platform	Key capability	Power	Approx. cost	Assessment
Raspberry Pi 4B	Native Camera Module 3 support and sufficient ML performance	Moderate	R0*	Selected: sufficient performance, lower energy demand and no acquisition cost.
Raspberry Pi 5 4GB	Faster CPU and greater processing headroom	High	R2 845 [47]	Extra performance did not justify its higher cost and power demand.
NVIDIA Jetson Orin Nano Super	GPU acceleration providing up to 67 TOPS	Very high	R8 669 [48]	Powerful, but well above the budget and unsuitable for solar operation.
ESP32-S3	Low-power microcontroller with wireless connectivity	Very low	R200 [49]	Unable to support the required camera, video and Linux-based ML stack.

*The Raspberry Pi 4B was supplied at no project cost. A comparable 4GB board retailed for approximately R1 639 [50]. Prices include VAT and were accessed on 16 September 2026.

Camera Module Introduce the Camera Module 3 as the standard camera module for RPis. Mention the other types of camera modules and then explain why the basic Camera Module 3 was eventually chosen:

- It has ample resolution and framerate for the task at hand
- It doesn't have an IR filter, which helps identify the iridescent sunbirds much more easily
- It is the most affordable camera module in the family
- It has a variable focus, which the cheaper Module 2 does not

Talk about how the more expensive modules were unnecessarily complicating the objective at a higher price The candidate cameras were compared according to image quality, focusing capability, cost and integration requirements, as summarised in Table 3.2.

Table 3.2: Comparison of candidate camera modules

Camera	Imaging capability	Focus	Approx. cost	Assessment
Camera Module 3	12 MP, HDR	Autofocus	R0*	Selected: best balance of image quality, size, compatibility and cost.
Camera Module 2	8 MP	Fixed	R290 [51]	Lower resolution and limited focusing flexibility.
HQ Camera and lens	12.3 MP, adjustable optics	Manual	R1 950 [52, 53]	Greater optical control, but bulky and unnecessarily expensive.
Raspberry Pi AI Camera	12 MP, on-sensor inference	Manual	R1 325 [54]	Onboard inference adds cost and deployment complexity.

*The Camera Module 3 was provided at no cost; its retail price was approximately R505 [55]. Prices include VAT.

Motion Sensor Introduce the VL53L1X ToF sensor as the chosen motion sensor and justify its use over others because:

- Programmable FOV, which decreases risk of background disturbances triggering it
- Interrupt output for RPi triggering without camera monitoring
- 50Hz measurement rate. Very fast response time
- ToF sensors enable birds to remain detected while stationary and feeding
- Its infrared emitter is eye-safe
- Ideal for short-range detection, with an adjustable threshold

Candidate visit-detection sensors were compared according to their detection method, sensing region, cost and suitability for monitoring the feeder port, as shown in Table 3.3.

Solar Power System Waveshare Model B power management system with MP 133x110 2.4W panel

- Wide input range of solar voltage: 6-24V
- Protection against overcharging, over-discharging, excessive current, overheating and reversed solar-panel polarity
- USB-C charging capability in case of insufficient solar energy input

Table 3.3: Comparison of candidate visit-detection sensors

Sensor	Detection method	Detection region	Approx. cost	Assessment
VL53L1X	Optical time-of-flight	Up to 4 m; adjustable region	R220 [56]	Selected: precise detection zone and reliable stationary-object detection.
HC-SR501	Passive infrared	Wide, adjustable cone	R22 [57]	Low power, but detects thermal motion rather than feeder occupancy.
HC-SR04	Ultrasonic ranging	0.02–4 m cone	R37 [58]	Low cost, but less directional and sensitive to outdoor conditions.
VL53L0X	Optical time-of-flight	Up to 2 m; fixed region	R40 [59]	Cheaper, but offers less range and detection-region control.

- 5V/3A output matches RPi power input requirements

Housing 3D printed with white PLA because:

- Cheap and easily accessible thanks to the university’s printing lab.
- Ideal for frequently-evolving prototypes that require quick production.
- Allows for an entirely customisable shape and features as required by the project.
- White colour specifically to minimise the heat absorbed from the sun and helps keep electronics cool.

3.5.2 Software Design Decisions

Operating System Introduce RPi Lite x64 as the selected OS, among several candidate OSs for the RPi 4B. It was selected because:

- It is headless, which meant it is one of the least computationally and energy-intensive OSs since there are no unneeded applications etc running
- It is 64-bit, which makes it compatible for use with all the libraries used by the classification algorithm. 32-bit would not have worked

The correct term is Raspberry Pi OS Lite (64-bit/ARM64) rather than x64, which normally refers to the x86-64 architecture. The candidates were compared according to resource demand, hardware support and software compatibility, as summarised in Table 3.4.

3.6. CONCLUSION

Table 3.4: Comparison of candidate operating systems

Operating system	Interface	Demand	Project compatibility	Assessment
Raspberry Pi OS Lite 64-bit [60,61]	Headless	Low	Native camera and GPIO support; compatible with ARM64 ML software	Selected: provides the required functionality with minimal overhead.
Raspberry Pi OS Lite 32-bit [60]	Headless	Low	Native hardware support, but limited to ARM32 software	Suitable for legacy software, but offers less ML compatibility and performance headroom.
Raspberry Pi OS Desktop 64-bit [60]	Desktop GUI	Higher	Same Raspberry Pi hardware support	The graphical desktop unnecessarily consumes storage, memory and processing resources.
Ubuntu Server 64-bit [62]	Headless	Moderate	Supports the Pi 4B, but requires more Pi-specific configuration	Capable, but less directly aligned with the official camera software stack.

Classification Model EfficientNetV2 was chosen as the convolutional backbone of the classification model, pretrained on the ImageNet dataset.

Dataset Mention licensing and their corresponding usage limitations.

Web Page

3.6 Conclusion

Chapter 4

Design

4.1 Introduction

The purpose of this chapter will be to present, in detail, the design of a smart sunbird feeder. The design explained in this chapter will build on the information presented in the previous chapters. The design approach used for this system follows the methodology and design decisions established in Chapter 3. As this system comprises of several hardware and software subsystems, this chapter will describe each subsystem in its own subsection.

4.2 Hardware Design

This section details the design of the physical/material subsystems and components of which the system is comprised. Overall, the subsystems comprising of the system's hardware are connected to each other through electrical and communication interfaces. The interaction between all of the hardware subsystems is represented in Figure 4.1, which shows the power and signal connections between each of the components. The hardware design of the system is presented in the following subsections, which include the feeder and mechanical design, embedded computing and camera, visit-detection hardware, storage and hardware interfaces, power and solar system, and enclosure and outdoor installation.

4.2.1 Feeder and Mechanical Design

As previously mentioned in Chapter 2 and Chapter 3, sunbirds are nectar-feeding birds. The proper storage and maintenance of a nectar feeder are key considerations when selecting the feeding mechanism. The nectar is required to be replaced every 48 hours at most, with its container being cleaned at similar intervals to remain hygienic and safe for bird use. These requirements translate to a need for an easily-accessible feeder that is robust to environmental factors and is capable of storing liquid reliably and without risk

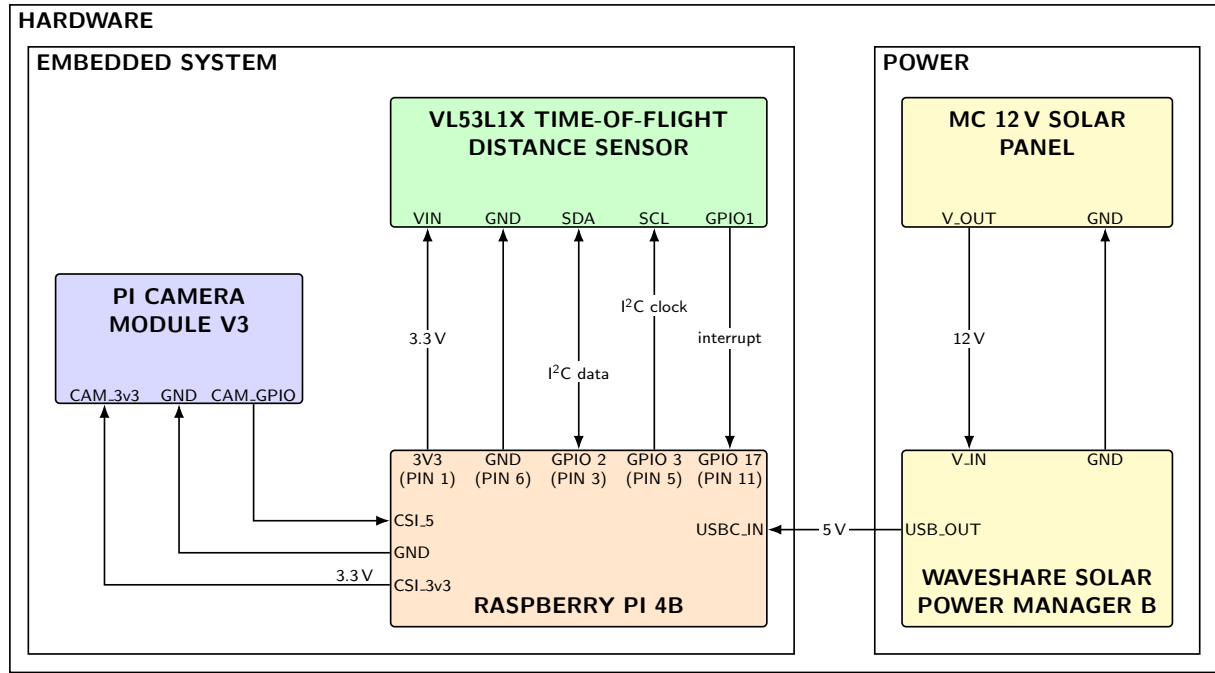


Figure 4.1: Hardware interfacing diagram of the smart sunbird feeder, showing the hardware subsystems and the power and signal connections between them.

of contamination.

Due to these intricate requirements of the feeder for this project, it was decided that this project will utilize a very simple, commercially available nectar feeder. The feeder selected for this project was chosen for its simplicity, availability, cost-effectiveness and proven satisfaction of all necessary concerns of reliability and safety. The feeder used for this project, a 750 ml sunbird nectar feeder from Stodols [63], is shown in Figure 4.2.



Figure 4.2: The commercially available Stodols 750 ml sunbird nectar feeder used in this project [63].

This project required multiple additional components to be physically connected to the

feeder. The stock feeder selected for this project offered little purchase for such components to be attached between its smooth glass body and minimal chain support mechanism. For this reason, it was decided to remove the chains entirely from the glass bottle and integrate the bottle into a larger housing that better accommodated for the introduction of the extra components required. The design of this housing is documented further in Section [4.2.6](#).

4.2.2 Embedded Computing and Camera

- Present the selected embedded computing platform.
- Present the selected camera module.
- Describe the relevant interfaces between the camera and embedded computer.
- Include important technical specifications relevant to system operation.
- Explain how the selected hardware satisfies the processing and imaging requirements of the project.

4.2.3 Visit-Detection Hardware

- Describe any sensor hardware used to detect or trigger bird visits.
- Explain sensor positioning and connection to the embedded computing platform.
- Discuss relevant detection range, response and interface considerations.
- If visit detection is implemented entirely in software, move the relevant discussion to Section [4.3](#).

4.2.4 Storage and Hardware Interfaces

- Describe the local storage hardware used by the system.
- Present the main electrical and communication interfaces between components.
- Include a simplified hardware connection diagram where appropriate.
- Discuss any supporting electronics required for system operation.

4.2.5 Power and Solar System

- Present the overall power architecture.
- Describe the battery, solar panel, charge controller and voltage-regulation components.
- Explain power distribution to the embedded computer and peripherals.
- Include provision for manual charging where applicable.
- Present relevant sizing calculations or design equations where necessary.

4.2.6 Enclosure and Outdoor Installation

- Describe the enclosure used for the electronics and power components.
- Explain protection against rain, moisture, dust and direct environmental exposure.
- Discuss ventilation and thermal considerations.
- Describe the mounting arrangement for the electronics, battery and solar panel.

4.3 Software Design

4.3.1 Software Architecture

- Present the overall software architecture.
- Include a software block diagram or flow diagram.
- Identify the main software modules and their interactions.

4.3.2 Visit Detection and Video Capture

- Describe the software logic used to detect or respond to a bird visit.
- Explain how video recording is initiated and terminated.
- Describe how recorded visits are separated and stored.
- Explain the handling of repeated or false triggers where relevant.

4.3.3 Video Processing and Species Identification

- Describe how recorded video is processed prior to inference.
- Explain the frame-selection method.
- Describe any image preprocessing.
- Present the selected species-classification model.
- Define the locally relevant classification classes.
- Explain inference and confidence handling.
- Describe how multiple frame predictions are combined into a visit-level result where applicable.

4.3.4 Data Logging and Storage

- Describe how visit information is logged.
- Present the database structure at an appropriate level of detail.
- Describe the storage of:
 - Date and time.
 - Video file location.
 - Predicted species.
 - Classification confidence.
 - Visit duration.
 - Processing status.
 - Relevant system logs.

4.3.5 Web Interface

- Describe the local web-interface architecture.
- Explain interaction between the web server, database and stored video files.
- Present the main interface functions, including:
 - Dashboard.

- Recent visit history.
- Species-identification results.
- Video playback.
- Basic system status.
- Relevant settings where implemented.

4.3.6 System Control and Reliability

- Describe software mechanisms used to support reliable unattended operation.
- Discuss relevant features such as:
 - Automatic startup.
 - Error handling.
 - Process recovery.
 - Storage management.
 - System logging.
 - Power-aware operation.
- Omit this subsection if these functions are more appropriately discussed within the preceding software subsections.

4.4 System Integration

- Explain how the hardware, software, mechanical and power subsystems are combined.
- Present the final integrated system architecture.
- Include photographs or annotated diagrams of the completed system where appropriate.
- Discuss any important integration constraints or modifications.
- Avoid repeating detailed subsystem descriptions already presented earlier in the chapter.

4.5 Bill of Materials

- Present the main components used in the completed system.
- Include:
 - Component.
 - Quantity.
 - Function.
 - Supplier or source.
 - Unit cost.
 - Total cost.
- Group components into mechanical, electronic and power categories where useful.

4.6 Conclusion

- Summarise the completed system design.
- Reinforce how the hardware and software subsystems combine into a functional smart sunbird feeder.
- Relate the final design back to the main system requirements.
- Transition to the subsequent chapter covering system testing, results and evaluation.

Chapter 5

Results

5.1 Results

- Describe the experimental setup (ie. Hardware)
- Describe your experiments
- Describe your results

This chapter should generally present your finding in the form of figures, tables, equations, code listings, etc. It is for presenting and discussing your findings, which can split into sections if the experiment has multiple parts, or stages. You might also want to have a ‘Discussion’ or ‘Discussion of Results’ chapter, which may focus on either more detailed discussion of particular results, or more comprehensive discussion of the results and system performance as a whole. If you have a Discussion of Results section, then you do not want to discuss too much about the specific results in this chapter and rather move the main discussion or reflective considerations of these to the Discussion of Results chapter.

However, and this is an important reminder, ensure that you do have some text of discussion for your results, to take the reader through your results and the figures you may provide. Make sure not to just put one figure after another without any attempt to explain the sequencing and what is being shown and perhaps some key issues in the figures presented (a monolithic sequence of figure after figure without any attempt at explanation will get undesirable responses from examiners).

If you have an experiments section, it is often useful to have a clear connection of experiment section corresponding to a result section showing results for that test. Examiners often appreciate that sort of clear and consistent structure that is easy to follow. For example, if you have section 5.1 as Modular Testing, then you could correspond to 6.1 Modular Testing Results (the two sections should be cross-linked with `\label` and `\ref`, in both directions)

5.2 Tips for Results Figures

Include good quality graphs (see Fig. 5.1 for an example).

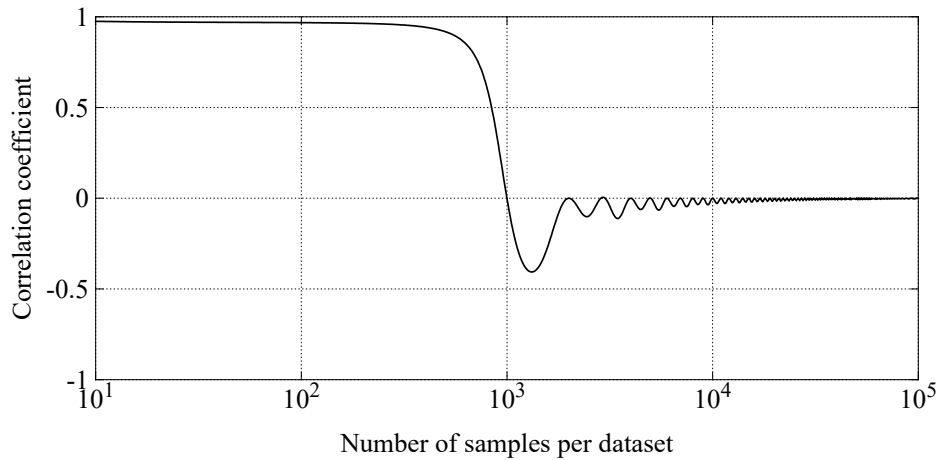


Figure 5.1: The correlation coefficient as a function of sample count

An easy way to obtain more space for an article is to use wide, flat figures, such as Fig. 5.2.

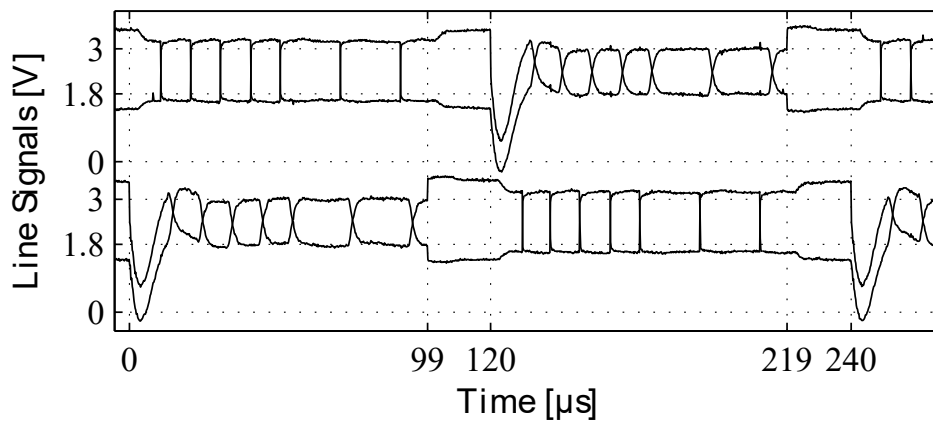


Figure 5.2: Oscilloscope measurement showing physical line signals on both ends of a transmission line during master switch-over

Always remember to include axes text, units and a meaningful caption in your graphs. When typing units, a μ sign has a tail! The letter “u” is not a valid unit prefix. When typing resistor values, use the Ω symbol.

5.3 Pictures and screenshots

When you include screenshots, pdfL^AT_EX supports JPG and PNG file formats. PNG is preferred for screenshots, as it is a loss-less format. JPG is preferred for photos, as it

results in a smaller file size. It's generally a good idea to resize photos (not screenshots) to be no more than 300 dpi, in order to reduce file size. **Never change the aspect ratio of screenshots and pictures!**

Fig. 5.3 shows an example of a PDF image with custom scaling.



Figure 5.3: An example image with custom scaling

Make sure to always use the best quality image possible. Use JPEG for photos, PNG for screenshots and PDF (scalable vector graphics) for everything else. JPEG is lossy, but good for photos, whereas PNG is lossless and good for images with large areas of solid colour.



(a) JPEG



(b) PNG



(c) SVG

Chapter 6

Conclusions

6.1 Conclusions

- Restate the problem. State the novel solution.
- Summarize what has been accomplished
- Summarize any limitations
- What worked really well and has a big impact?

6.2 Future Work

- How do you hope to continue work on this topic?
- Are there possible extensions?
- What are some improvements that could be made?

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Appendix A

Supporting Data

Appendix sections are where you can place large figures, data tables, and spinets of code. Use appendices to your benefit to keep the body of your thesis concise!

The lyrics found below are for your enjoyment, but also serve an important role in demonstrating latex syntax for formatting text and in-text citations.

A.1 Lyrics to Soft Kitty

Soft kitty, warm kitty
Little ball of fur
Happy kitty, sleepy kitty
Purr, purr, purr

This has been brought to you by Sheldon Cooper.

Appendix B

Satirical Support

This section provides some comic relief. In addition, it serves as an example of how to insert an image into your thesis with proper caption, label and citation.



Figure B.1: You must prepare to defend your thesis.